



0. Data Analytics

Turning data into knowledge, and knowledge into better decisions.

Contents:

[1. What is Data Analytics?](#)

[2. What is Data?](#)

[3. What is a Dataset?](#)

[4. Why is Data Analytics Important?](#)

[5. Data Visualization in Analytics](#)

[6. The Data Analytics Process](#)

[7. Tools for Data Analytics](#)

[Summary](#)

[References](#)

Everywhere we look today, people are talking about data. Companies collect data. Governments collect data. Apps collect data. Even the watch on your wrist is probably collecting data.

Because of that, terms like **Data Analytics**, **Data Analysis**, **Business Analytics**, and even **Data Science** are often used interchangeably. Technically, these fields have different scopes, but they all revolve around the same central idea: extracting useful information from data.

Throughout this course, we will simply use the term **Data Analytics**. More importantly, this course is not about memorizing Python libraries or creating colorful charts. Those are just tools. Before we learn how to analyze data, we first need to understand what data actually is, why it matters, and what questions we should be asking.

So before we start using Python or building visualizations, let's answer a few fundamental questions.

1. What is Data Analytics?

Let's begin with the most important question: ***What exactly is Data Analytics?***

Data Analytics is the process of collecting, organizing, transforming, and analyzing data to discover patterns, answer questions, and support better decision-making.

Notice that analysis is only one part of the process. Before we can discover anything useful, we usually need to collect the data, organize it into a suitable format, clean any inconsistencies, and prepare it for analysis. Only then can we start looking for trends, relationships, and insights.

Ultimately, the goal is simple: **Turn raw data into information that helps us make better decisions.**

2. What is Data?

Now that we know what Data Analytics does, we should ask another question: ***What exactly is data?*** Data is a collection of raw facts, measurements, observations, or records that describe something.

By itself, data usually does not tell us very much. It simply records what happened.

Data can take many different forms:

- Numbers such as sales, temperatures, or exam scores.
- Text such as customer reviews or emails.
- Images, videos, and audio recordings.
- Dates, locations, and user activities.

We collect data from many different sources, including sensors, websites, mobile applications, surveys, and databases. It can be stored in spreadsheets, CSV files, SQL databases, cloud storage, and many other formats.

Think of data as the raw ingredients in a kitchen. On their own, the ingredients are not the final meal. It is only after we organize, prepare, and combine them that something useful begins to appear. The same idea applies to Data Analytics.

3. What is a Dataset?

Individual pieces of data are useful, but they become much more valuable when they are organized together. This organized collection is called a **dataset**.

Most datasets are arranged as a table consisting of rows and columns.

- Each **row** represents one observation or record.
- Each **column** represents one attribute or feature describing that record.

For example, imagine a dataset called **Student**. Each row represents one student, while the columns might contain information such as: Name; Age; Gender; Marks. As the number of students grows, the dataset simply gains more rows while keeping the same structure.

Real-world datasets come in many forms. Some famous examples include the ***Iris Dataset, Wine Dataset, Weather Dataset, Medical Dataset, Online Shopping Dataset***, and countless others collected by companies and researchers every day.

Throughout this course, our football dataset follows exactly the same principle. Every row represents one player, while every column describes one of that player's attributes.

4. Why is Data Analytics Important?

Now that we understand what data and datasets are, another question naturally follows: ***Why do we spend so much effort analyzing data?***

The answer is simple. Raw data rarely provides useful answers by itself. Its value comes from what we can learn after analyzing it. Data Analytics helps us do several important things:

- **Convert raw data into meaningful information:** Real-world data is often incomplete, inconsistent, or messy. Analytics helps us clean, organize, and interpret it so we can discover useful relationships.
- **Identify patterns and trends:** Large datasets often contain behaviors that are impossible to notice by simply looking at individual records. Analytics helps reveal long-term trends, hidden relationships, and unexpected discoveries.

- **Make predictions:** Historical data allows us to estimate future outcomes. Businesses forecast sales, hospitals predict patient risks, banks detect fraud, and sports teams evaluate player performance using predictive analytics.
- **Support better decisions:** Instead of relying entirely on intuition, organizations can make decisions supported by evidence. Better information usually leads to better planning, lower costs, and improved performance.

In other words, Data Analytics helps us replace guesses with evidence.

5. Data Visualization in Analytics

Once we have collected, cleaned, and analyzed our data, we still face one important challenge: ***How do we communicate what we found?***

This is where **Data Visualization** becomes important. Visualization transforms tables of numbers into charts, graphs, and dashboards that people can understand much more quickly.

A well-designed visualization allows us to spot trends, compare values, detect outliers, and explain our findings far more effectively than pages of raw numbers.

Although Data Visualization is an important part of Data Analytics, the two are not exactly the same.

- Data Analytics focuses on preparing, processing, analyzing, and interpreting data.
- Data Visualization focuses on presenting those results clearly and effectively.

Throughout this course, we will use visualization as a tool to better understand our analyses rather than simply creating attractive charts.

6. The Data Analytics Process

Up until now, we have talked about what Data Analytics is and why it matters. But how do we actually perform Data Analytics?

Many beginners imagine opening a dataset, drawing a few charts, and calling it analysis. In reality, analytics is a process. Each step builds upon the previous one, and skipping one often leads to unreliable results.

Let's walk through a typical Data Analytics workflow.

- **Data Collection:** Everything begins with data. Before we can analyze anything, we first need to gather it from one or more sources.

Common sources include:

- CSV files
- Excel spreadsheets
- JSON files
- SQL databases
- NoSQL databases such as MongoDB
- APIs, sensors, websites, and many others

Without data, there is nothing to analyze.

- **Data Cleaning:** Unfortunately, real-world data is rarely perfect. Datasets often contain missing values, duplicate records, incorrect formats, impossible values, or irrelevant features. Before analysis, we need to clean these problems.

Typical cleaning tasks include:

- Handling missing values
- Removing duplicate records
- Detecting outliers
- Removing redundant features
- Correcting inconsistent data formats

This is often the most time-consuming part of a data analytics project, but it is also one of the most important. Poor-quality data almost always leads to poor-quality analysis.

- **Data Transformation:** Once the data is clean, we often need to prepare it for analysis. This step transforms the data into a more useful format.

Common transformations include:

- Normalizing numerical values
- Encoding categorical variables into numerical values
- Creating derived variables from existing features
- Grouping continuous values into categories

The goal is not to change the meaning of the data, but to represent it in a form that is easier to analyze.

- **Data Exploration:** Now we can finally begin asking questions. Exploratory Data Analysis (EDA) helps us understand the dataset before drawing conclusions or building predictive models. We summarize the data using descriptive statistics and simple visualizations.

Some common techniques include:

- Mean
- Median
- Mode
- Standard deviation
- Covariance Matrix
- Correlation Matrix

At this stage, we are not trying to prove anything. We are simply becoming familiar with our data and looking for interesting patterns.

- **Data Visualization:** Numbers alone can be difficult to interpret. Visualization allows us to transform tables into charts that reveal patterns much more clearly.

Common visualizations include:

- Bar charts
- Line charts

- Scatter plots
- Histograms
- Box plots
- Heatmaps

Good visualizations help us communicate findings and often reveal relationships that would otherwise remain hidden.

- **Interpretation and Reporting:** Finding patterns is not the final goal. We still need to explain what those patterns actually mean.
 - This stage converts analysis into useful insights through reports, dashboards, presentations, and recommendations that support decision-making.

After all, a great analysis that nobody understands has very little value.

- **Model Training:** Sometimes our goal is not only to understand the past, but also to predict the future. In these cases, we can train Machine Learning models for tasks such as prediction, classification, clustering, or recommendation.
 - This step is optional. Many Data Analytics projects end after reporting insights and never require Machine Learning.

Throughout this course, we will spend most of our time on the core stages of analytics:

Data Cleaning → Data Transformation → Data Exploration → Data Visualization

These four stages form the foundation of almost every data analytics project.

7. Tools for Data Analytics

Now that we understand the workflow, we need tools to perform each step efficiently. And for your information, there is no single tool that does everything perfectly. Different tools excel at different tasks.

Some of the most popular tools include:

- **Microsoft Excel** for quick analysis and spreadsheets.
- **Microsoft SQL Server** (and SQL in general) for querying and managing databases.
- **Python**, especially the **Pandas** library, for data cleaning, transformation, analysis, and visualization. Throughout this course, we will primarily use Python in Google Colab.
- **Power BI** for interactive dashboards and business intelligence.
- Many other tools such as Tableau, R, Apache Spark, and cloud analytics platforms.

Of course, tools alone are not enough. We also need data to analyze.

Fortunately, there are many public sources where we can obtain datasets for learning, research, and real-world projects.

Some of the most popular sources include:

- **Kaggle** – One of the largest collections of public datasets, ranging from sports and finance to healthcare, computer vision, and natural language processing.
- **UCI Machine Learning Repository** – A classic collection of benchmark datasets widely used in machine learning research and education.
- **Data.gov** (and similar government open-data portals) – Public datasets covering population, transportation, healthcare, economics, weather, and many other topics.
- **GitHub** – Many developers and researchers publish datasets alongside their projects, making GitHub another valuable source of open data.
- **Built-in datasets from Python libraries** – Libraries such as **Scikit-learn**, **Seaborn**, and **Statsmodels** include several toy datasets like Iris, Wine, Breast Cancer, Titanic, and Tips. These datasets are small, clean, and perfect for learning new concepts before moving on to larger real-world datasets.

Throughout this course, we will use a mixture of **randomly-generated datasets** (mainly for the analytics part), **toy datasets** and **larger real-world datasets** (mainly for the visualization part). Toy datasets help us understand new

concepts quickly, while real datasets expose us to the kinds of challenges encountered in practical data analytics.

Rather than trying to learn every tool or memorize every dataset source, we will focus on understanding the analytics process itself. Once we understand the process, learning a new tool, or working with a new dataset, becomes much easier.

Summary

Data Analytics is much broader than simply creating charts or calculating statistics.

It is a complete process that begins with collecting data, continues through cleaning, transforming, exploring, and visualizing it, and finally turns that data into insights that support better decisions. Sometimes, the journey continues into Machine Learning to build predictive models, but many successful analytics projects never require one.

Throughout this course, our focus will be on developing the skills needed to understand data, ask meaningful questions, and communicate findings effectively. The tools may change over time, but the process of thinking critically about data remains the same.

References

<https://www.geeksforgeeks.org/data-analysis/what-is-data-analytics/>

<https://www.tableau.com/analytics/what-is-data-analytics>

Next Section:

 [1.1. Imputation of Missing Values](#)